

Experiment Number:03

Experiment Name: Simplify function & implement them by logic gates

Theory:

Karnaugh Map (K-Map) is a graphical method used to simplify Boolean expressions. It arranges the truth-table values in Gray code order, allowing adjacent 1s to be grouped together. By forming groups of 1, 2, 4, 8, etc., redundant variables can be eliminated and a simpler Boolean expression can be obtained. This reduces the number of logic gates required for circuit implementation.

After simplifying the Boolean function using a K-Map, the resulting expression can be implemented using logic gates such as AND, OR, and NOT gates in IC circuits. Each product term is implemented using AND gates, complemented variables are generated using NOT gates, and the product terms are combined using an OR gate. Thus, K-Map simplification helps design a circuit that is simpler, faster, and uses fewer hardware components.

IC Circuits Diagrams:

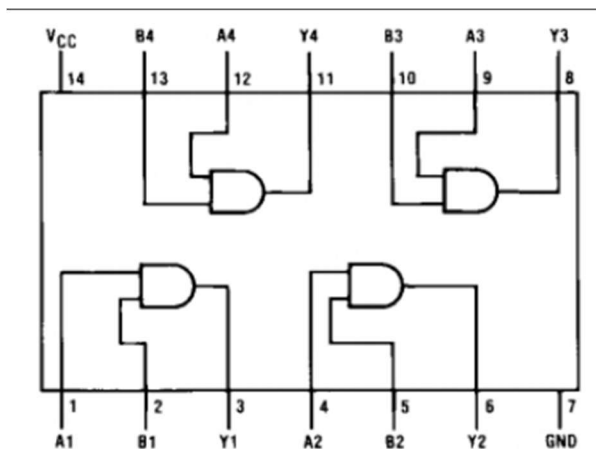


Fig 01:AND gate 7408

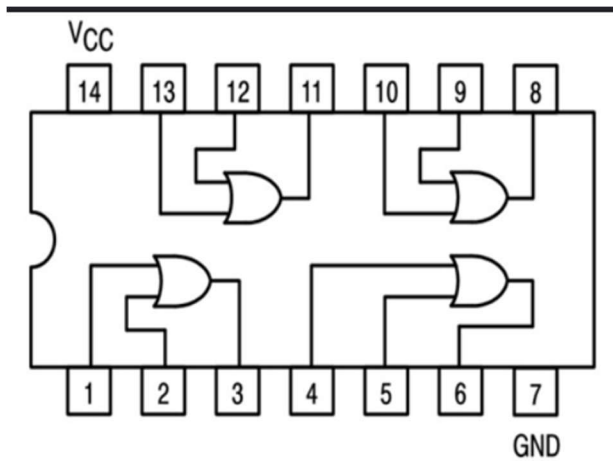


Fig 02: OR gate 7432

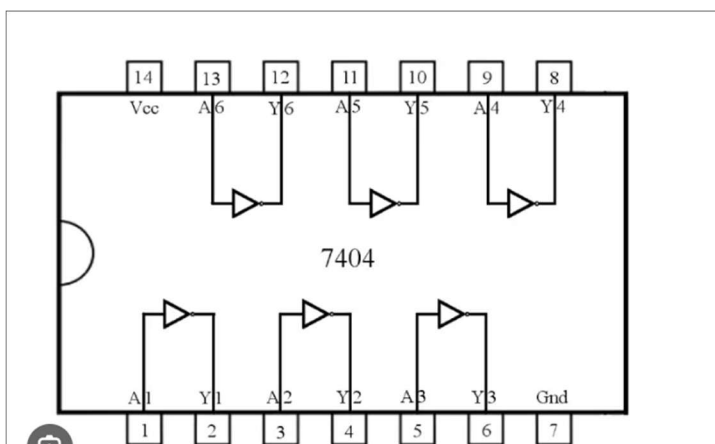


Fig 03: NOT gate 7404

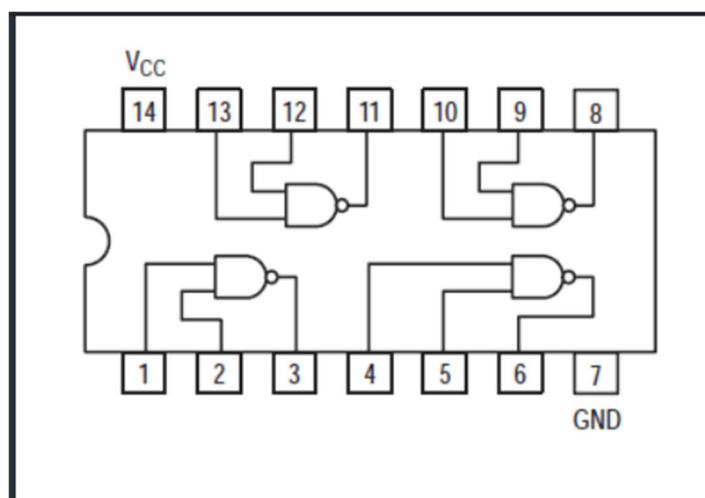


Fig 04: NAND gate 7400

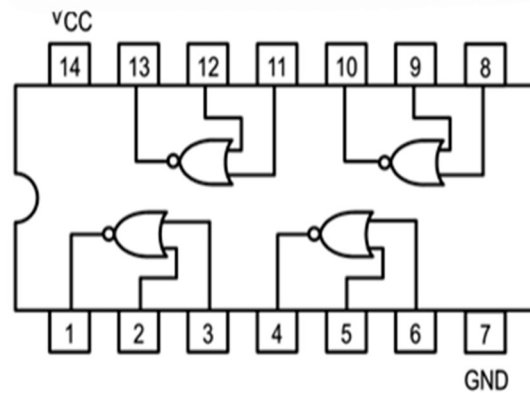


Fig 05:NOR gate 7402

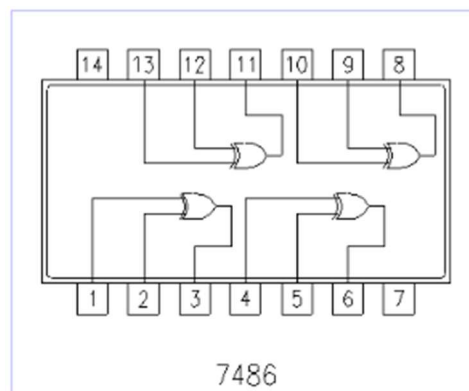


Fig 06:XOR gate 7486

Function: $F(A,B,C)=A'BC+AB'C+ABC'+ABC$

Implementations:

1. $F(A,B,C)=A'BC+AB'C+ABC'+ABC$

K-MAP:

A\BC	B'C'	B'C	BC	BC'
A'	0	0	1	0
A	0	1	1	1

Simplified version: $AB+BC+CA$

Implementation with IC Circuits:

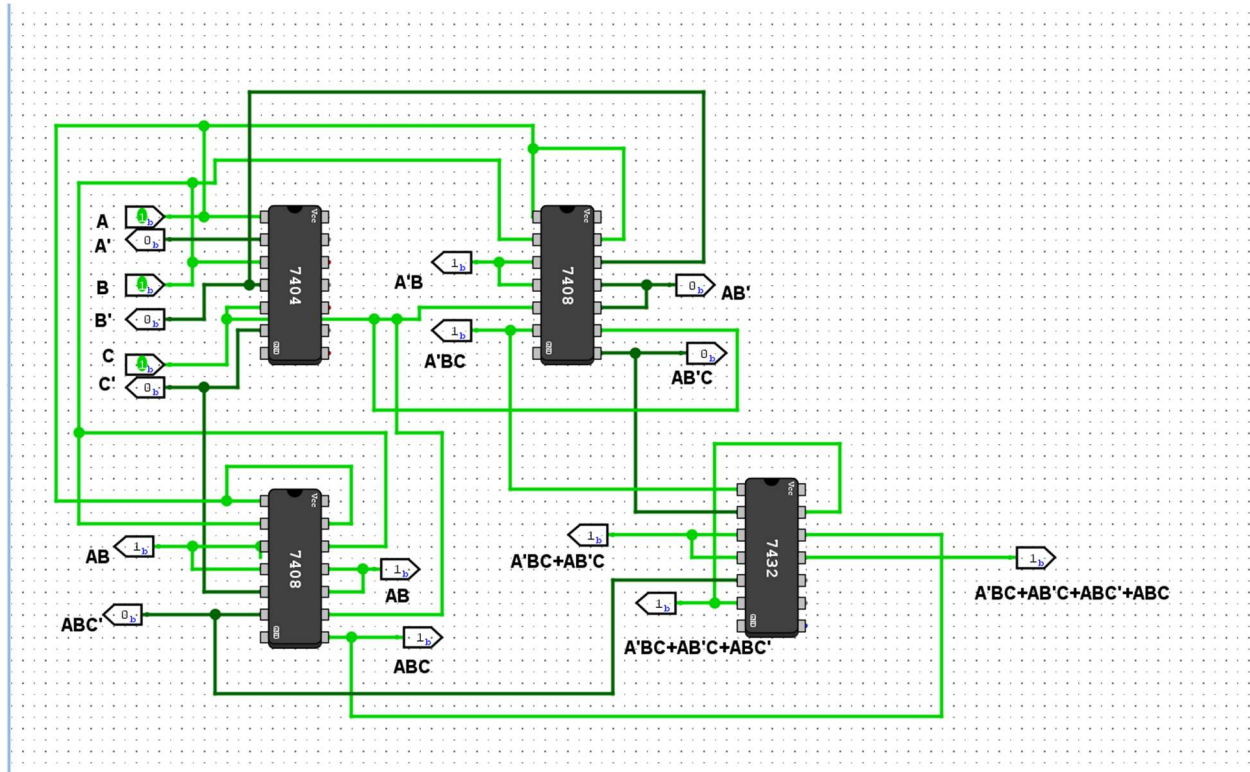


Fig 07: Implementation with IC circuits at Logisim